

* Principal component Analysis :-

- Results from Linear Algebra :-

① Suppose $\Sigma_{p \times p}$ is a symmetric matrix then Σ has real eigenvalues and the eigen vectors corresponding to different eigenvalues are orthogonal.

$\lambda_1, \lambda_2, \dots, \lambda_p$ be the eigen values of Σ
 & $\underline{e}_1, \underline{e}_2, \dots, \underline{e}_p$ be the corresponding eigen vectors
 then $\underline{e}_i \perp \underline{e}_j \quad \forall i \neq j \in \{1, 2, \dots, p\}$
 i.e. $\boxed{\underline{e}_i^T \underline{e}_j = 0}$

② Spectral decomposition of Σ (symmetric matrix) exists.

$$\Sigma_{p \times p} = V_{p \times p} \Lambda_{p \times p} V_{p \times p}^T$$

where Λ is a diagonal matrix. ; $\Lambda = \text{diag}(\quad)$
 V is a (symmetric) orthogonal matrix. ; $\boxed{V V^T = V^T V = I_p}$

Also $\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$

then j th column of V matrix is the eigen vector corresponding to $\boxed{\lambda_j}$

$$V = [\underline{v}_1, \underline{v}_2, \dots, \underline{v}_p] \Rightarrow (\lambda_j, \underline{v}_j) \quad \left[\begin{array}{l} \text{from result ①} \\ \underline{v}_i^T \underline{v}_j = \delta_{ij} \end{array} \right]$$

$$\Sigma = \left[\begin{array}{c} \underline{v}_1 \\ \vdots \\ \underline{v}_p \end{array} \right]_{p \times p} \left[\begin{array}{ccc} \lambda_1 & & \\ & \lambda_2 & \\ & & \ddots \\ & & & \lambda_p \end{array} \right] \left[\begin{array}{c} \underline{v}_1^T \\ \underline{v}_2^T \\ \vdots \\ \underline{v}_p^T \end{array} \right] \rightarrow p \times p$$

$$\Sigma = \sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i^T$$

$$\Rightarrow \sum_{p \times p} \underline{v}_j = \sum_{p \times p} \sum_{i=1}^p \lambda_i \underbrace{\underline{v}_i}_{p \times 1} \underbrace{\underline{v}_i^T}_{1 \times p} \underbrace{\underline{v}_j}_{p \times 1} = \boxed{\lambda_j \underline{v}_j}_{p \times 1} \quad \left(\underline{v}_i^T \underline{v}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases} \right)$$

② without loss of generality, we can assume that

$$\boxed{\lambda_1, \lambda_2, \dots, \lambda_p}$$

$$\Sigma = \sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i^T \quad (\Sigma \text{ is Symmetric})$$

$$= V \Lambda V^T$$

③ Suppose Σ is Symmetric matrix, then

$$\sup_{\underline{u} \in \mathbb{R}^p} \frac{\underline{u}^T \Sigma \underline{u}}{\underline{u}^T \underline{u}} = \lambda_1, \text{ where } \lambda_1 \text{ is the largest eigen value of } \Sigma.$$

Proof:

$$\boxed{\Sigma = V \Lambda V^T}$$

$$\begin{aligned} \frac{\underline{u}^T \Sigma \underline{u}}{\underline{u}^T \underline{u}} &= \frac{\underline{u}^T (V \Lambda V^T) \underline{u}}{\underline{u}^T \underline{u}} = \frac{\underline{u}^T \left(\sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i^T \right) \underline{u}}{\underline{u}^T \underline{u}} \\ &= \frac{\sum_{i=1}^p \lambda_i \underline{u}^T \underline{v}_i \underline{v}_i^T \underline{u}}{\underline{u}^T \underline{u}} = \frac{\sum_{i=1}^p \lambda_i \omega_i^2}{\underline{u}^T \underline{u}} \quad \left[\omega_i = \underline{u}^T \underline{v}_i \right. \\ &\quad \left. (\text{Scalar}) \right] \\ &\leq \frac{\sum_{i=1}^p \lambda_1 \omega_i^2}{\underline{u}^T \underline{u}} = \lambda_1 \frac{\sum_{i=1}^p \omega_i^2}{\underline{u}^T \underline{u}} \quad \text{--- ①} \end{aligned}$$

Now, $\omega_i^2 (\text{Scalar}) = \omega_i \omega_i^T = \boxed{\omega_i^T \omega_i}$

$$\begin{aligned} \omega_i^T \omega_i &= \underline{v}_i^T \underline{u} \underline{u}^T \underline{v}_i \\ &= \text{tr}(\underline{u}^T \underline{u} \underline{v}_i^T \underline{v}_i) \\ &= \text{tr}(\underline{v}_i^T \underline{v}_i \underline{u}^T \underline{u}) \\ \frac{\sum \omega_i^2}{\underline{u}^T \underline{u}} &= \frac{\underline{u}^T \Sigma \underline{u}}{\underline{u}^T \underline{u}} = \frac{\underline{u}^T V \Lambda V^T \underline{u}}{\underline{u}^T \underline{u}} = \boxed{1} \end{aligned}$$

So substituting in ① we get $\boxed{\frac{\underline{u}^T \Sigma \underline{u}}{\underline{u}^T \underline{u}} \leq \lambda_1}$

Also

$$\begin{aligned} \frac{\underline{u}^T (V \Lambda V^T) \underline{u}}{\underline{u}^T \underline{u}} &\leq \frac{\underline{u}^T (V (\lambda_1 I_p) V^T) \underline{u}}{\underline{u}^T \underline{u}} \\ &= \lambda_1 \frac{\underline{u}^T (V V^T) \underline{u}}{\underline{u}^T \underline{u}} \quad \xrightarrow{I_p} \\ &= \lambda_1 \frac{\underline{u}^T \underline{u}}{\underline{u}^T \underline{u}} = \lambda_1 \end{aligned}$$

Hence

$$\boxed{\sup \left[\frac{\underline{u}^T (V \Lambda V^T) \underline{u}}{\underline{u}^T \underline{u}} \right] = \lambda_1}$$

Also

$$\inf \left[\frac{\underline{u}^T (V \Lambda V^T) \underline{u}}{\underline{u}^T \underline{u}} \right] = \lambda_p \quad (\text{where } \lambda_p \text{ is the smallest eigenvalue of } \Sigma)$$

* Principal components:

Suppose $\underline{X}_{p \times 1} = \begin{pmatrix} X_1 \\ X_2 \\ \vdots \\ X_p \end{pmatrix}$ is a Random Vector with mean i.e. $E(\underline{X}) = \underline{1}$ & $\text{cov}(\underline{X}) = \Sigma_{p \times p}$. Then,

PC's are uncorrelated linear combinations $\underline{Y} = (Y_1, Y_2, \dots, Y_p)'$ derived from $(X_1, X_2, \dots, X_p)$ with

$$\text{var}(Y_1) \gg \text{var}(Y_2) \gg \dots \gg \text{var}(Y_p)$$

$$\underline{X} \rightarrow \underline{Y} = V^T (\underline{X} - \underline{1}) \quad \text{with } \Sigma = V \Lambda V^T \text{ (Spectral decomposition)}$$

$$\Lambda = \text{diag}(\lambda_1, \lambda_2, \dots, \lambda_p)$$

with $\lambda_1 > \lambda_2 > \dots > \lambda_p$

and $(\lambda_1, \underline{v}_1), \dots, (\lambda_p, \underline{v}_p)$

be the eigenvalue-eigen vector pairs;

The transformation $\underline{X} \rightarrow \underline{Y}$ is one-one transformation.

$$\boxed{\underline{X} = \underline{1} + V \underline{Y}}$$

② Prove that Y_1 has the maximum variance among $Y_1, Y_2, \dots, Y_p$.

Let $\underline{Y} = (Y_1, Y_2, \dots, Y_p)'$ Given $\underline{X} = (X_1, X_2, \dots, X_p)'$

$$E(\underline{X}) = \underline{\mu} = (\mu_1, \dots, \mu_p)'$$

$$V(\underline{X}) = \Sigma = V \Lambda V^T$$

with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$

$$\lambda_1 > \lambda_2 > \dots > \lambda_p$$

$$\underline{V} = [\underline{v}_1, \dots, \underline{v}_p]$$

$$\begin{bmatrix} Y_1 \\ Y_2 \\ \vdots \\ Y_p \end{bmatrix} = \begin{bmatrix} \underline{v}_1' \\ \vdots \\ \underline{v}_p' \end{bmatrix} \begin{bmatrix} X_1 - \mu_1 \\ \vdots \\ X_p - \mu_p \end{bmatrix}$$

$p \times p$

Now $\text{Var}(\underline{Y}) = V^T \Sigma V = V^T V \Lambda V^T V$

$$= \begin{bmatrix} \underline{v}_1' \\ \vdots \\ \underline{v}_p' \end{bmatrix} [\underline{v}_1, \dots, \underline{v}_p] \text{diag}(\lambda_1, \dots, \lambda_p) \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix} \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}$$

we know

$$= \text{diag}(\lambda_1, \dots, \lambda_p) = \Lambda$$

$$V^T V = I_p$$

$$\text{Var}(\underline{Y}) = \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_p \end{bmatrix} \Rightarrow \boxed{\text{Var}(Y_i) = \lambda_i}$$

given $\lambda_1 > \lambda_2 > \dots > \lambda_p$

$$\Rightarrow \boxed{\text{Var}(Y_1) > \text{Var}(Y_2) > \dots > \text{Var}(Y_p)}$$

* Standardized Linear Combination (SLC):

An SLC of a Random Vector $\underline{X}_{p \times 1}$ is $\underline{e}'_{1 \times p} (\underline{X} - \underline{\mu})$

with $\|\underline{e}\| = 1$, $\boxed{\sum_{i=1}^p e_i^2 = 1}$

$$Y_1 = \underline{v}_1' (\underline{X} - \underline{\mu}) \Rightarrow \boxed{Y_1 \text{ is an SLC}}$$

* Y_1 has maximum variance among all SLC's of $\underline{X}$.

let Y^* be any other SLC of $\underline{X}$. i.e. $\boxed{Y^* = \underline{u}' (\underline{X} - \underline{\mu})}$

with $\boxed{\underline{u}' \underline{u} = 1}$

$$\text{var}(y^*) = \underline{u}' \text{var}(X) \underline{u} = \underline{u}' \Sigma \underline{u}$$

$$= \underline{u}' \underline{V} \underline{V}' \underline{u}$$

$$\leq \underline{u}' \underline{V} (\lambda \underline{I}_p) \underline{V}' \underline{u}$$

$$= \frac{\underline{u}' \Sigma \underline{u}}{\underline{u}' \underline{u}} \leq \lambda_1$$

$$\lambda_1 = \text{var}(Y_1)$$

Result: $\sup_{\underline{u} \in \mathbb{R}^p} \frac{\underline{u}' \Sigma \underline{u}}{\underline{u}' \underline{u}} \leq \lambda_1$

$$\Rightarrow \boxed{\text{var}(y^*) \leq \text{var}(Y_1)}$$

$\rightarrow$ If $\Sigma = I_p$ then $V(Y_i) = 1 \quad \forall i = 1, \dots, p$.

Result: Suppose $Z = \frac{\underline{u}^T (X - \underline{\mu})}{\|\underline{u}\|}$ and Z is uncorrelated to

$Y_1, Y_2, \dots, Y_k$ then Z has maximum variance if $\boxed{Z = Y_{k+1}}$

Proof:

we have $\text{cov}(Z, Y_j) = 0 \quad \forall j = 1, \dots, k$

$$\text{cov}(\underline{u}^T (X - \underline{\mu}), Y_j) = 0$$

$$\text{cov}(\underline{u}^T (X - \underline{\mu}), \underline{v}_j^T (X - \underline{\mu})) = 0$$

$$\Rightarrow \underline{u}' \underline{V} \underline{V}' \underline{u} = 0$$

$$\Rightarrow \underline{u}' \left(\sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i' \right) \underline{u} = 0 \quad \forall j = 1, \dots, k$$

$$\Rightarrow \sum_{i=1}^p \underline{u}' \lambda_i \underline{v}_i \underline{v}_i' \underline{u} = 0 \quad \left(\underline{u}' \underline{v}_j = \begin{cases} 0 & i \neq j \\ 1 & i = j \end{cases} \right)$$

$$\Rightarrow \underline{u}' \lambda_j \underline{v}_j = 0$$

$$\Rightarrow \boxed{\lambda_j \underline{u}' \underline{v}_j = 0 = \lambda_j \underline{v}_j^T \underline{u}} \quad \text{--- (1)}$$

$$\forall j = 1, \dots, k$$

$$\text{var}(Z) = \text{var}(\underline{u}' (X - \underline{\mu}))$$

$$= \underline{u}' \Sigma \underline{u}$$

$$= \underline{u}' \left[\sum_{i=1}^p \lambda_i \underline{v}_i \underline{v}_i' \right] \underline{u}$$

⑥

$$= \sum_{i=1}^p \lambda_i a^T u^T v_i v_i^T u$$

$$= \sum_{i=1}^k \lambda_i (u^T v_i)(v_i^T u) + \sum_{i=k+1}^p \lambda_i (u^T v_i)(v_i^T u)$$

from ① $\downarrow$
0

$$= \sum_{i=k+1}^p \lambda_i \omega_i^2 \quad (\omega_i = \underbrace{u^T}_{1 \times p} \underbrace{v_i}_{p \times 1}) \quad \forall i = k+1, \dots, p$$

$$\leq \lambda_{k+1} \sum_{i=k+1}^p \omega_i^2 \quad \text{--- ②}$$

now

$$\sum_{i=k+1}^p \omega_i^2 \leq \sum_{i=1}^p \omega_i^2 = \sum_{i=1}^p u^T v_i v_i^T u$$

$$= u^T \sum_{i=1}^p v_i v_i^T u \quad \left[\sum_{i=1}^p v_i v_i^T = V V^T = I_p \right]$$

$$= u^T u = 1$$

from ②

$$\lambda_{k+1} \sum_{i=k+1}^p \omega_i^2 \leq \lambda_{k+1}$$

Hence,

$$\boxed{\text{var}(z) \leq \lambda_{k+1}} \rightarrow \boxed{\max(\text{var}(z)) = \lambda_{k+1}}$$

Result:

Among the SLC of X , y_p has the least variance.

SLC: $a^T(X - \mu)$ where $\|a\| = 1$.

$$\text{var}(a^T(X - \mu)) \geq \lambda_p = \text{var}(y_p)$$

Proof:

LHS.

$$\text{var}(a^T(X - \mu)) = a^T \Sigma a$$

$$= a^T \sum_{i=1}^p \lambda_i v_i v_i^T a$$

$$= \sum_{i=1}^p \lambda_i a^T v_i v_i^T a = \sum_{i=1}^p \lambda_i \omega_i^2 \quad [\omega_i = a^T v_i]$$

$$\geq \lambda_p \sum_{i=1}^p \omega_i^2 = \lambda_p a^T \sum_{i=1}^p v_i v_i^T a = \boxed{\lambda_p} : (\min(\text{var}(a^T x)) = \lambda_p)$$

→ For a covariance matrix Σ , the total variation (TV) is defined as $\text{tr}(\Sigma)$.

$$\text{tr}(\Sigma) = \sum_{i=1}^p \lambda_i = \sum_{i=1}^p \text{var}(Y_i)$$

Total variation of $(Y_1, Y_2, \dots, Y_k) = \sum_{i=1}^k \lambda_i \quad (k \leq p)$

variability explained by the first 'k' P.C is

$$\frac{\sum_{i=1}^k \lambda_i}{\sum_{i=1}^p \lambda_i}$$

→ As the principal components do not have their own practical interpretation. It is a good exercise to see for the ~~variables~~ correlation between the true (original) variable and the ~~first~~ formed P.C. i.e $\text{Corr}(Y_k, X_j) \quad \forall i, j, k \in \{1, \dots, p\}$

$$\begin{aligned} \Rightarrow \text{Cov}(\underline{Y}, \underline{X}) &= \text{Cov}(\underline{V}^T(\underline{X} - \underline{\mu}), \underline{X}) \\ &= \underline{V}^T \Sigma = \underline{V}^T \underline{V} \underline{\Lambda} \underline{V}^T \\ &= \underline{\Lambda} \underline{V}^T \\ &= \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_p \end{bmatrix} \begin{bmatrix} u_{11} \\ \vdots \\ u_{p1} \end{bmatrix} \end{aligned}$$

$$\text{Cov}(Y_k, X_j) = \cancel{\lambda_k} u_{jk} \lambda_k u_{jk}$$

$$v(Y_k) = \lambda_k$$

$$v(X_j) = \sigma_{jj}$$

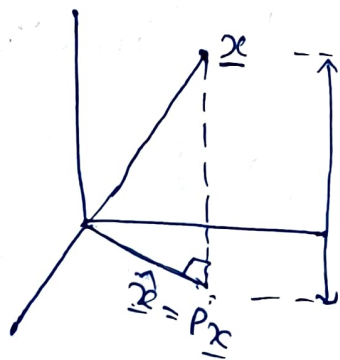
$$\Rightarrow \text{Corr}(Y_k, X_j) = \frac{\lambda_k u_{jk}}{\sqrt{\lambda_k} \sqrt{\sigma_{jj}}} = \boxed{u_{jk} \sqrt{\frac{\lambda_k}{\sigma_{jj}}} = \rho_{kj}}$$

$$\begin{aligned}
 \textcircled{8} \Rightarrow \sum_{k=1}^b \beta_{kj}^2 &= \sum_{k=1}^p \frac{\lambda_k y_{jk}^2}{\sigma_{jj}} \\
 &= \frac{1}{\sigma_{jj}} \sum_{k=1}^p \lambda_k y_{jk}^2 \\
 &= \frac{1}{\sigma_{jj}} \underbrace{\sum_{k=1}^p \lambda_k y_{jk} y_{jk}}_{\sigma_{jj}} \\
 &= \frac{\sigma_{jj}}{\sigma_{jj}} = 1
 \end{aligned}$$

β_{kj}^2 is interpreted as proportion of variability in X_j explained by X_k

Algebra Result:

- Suppose S is a subspace of $\mathbb{R}^p$ and $a \in \mathbb{R}^p$.
Then $a + S = \{a + x \mid x \in S\}$ is called an affine space.
- orthogonal projection of x onto S is Px where P is the orthogonal projection matrix.



- P satisfies $P = P^T$ and $P^2 = P$. (Symmetric & idempotent)
- Px is orthogonal projection of x onto S .
- $\hat{x} = a + P(x - a)$ is the orthogonal projection of x onto $a + S$.
- Given x , and for an affine space $a + S$, we write.

$$d(a, S) = E(\|x - \hat{x}\|^2)$$

Theorem: Among all k -dimensional subspaces of $\mathbb{R}^p$ and all vectors $\underline{a} \in \mathbb{R}^p$, $d(\underline{a}+S, \underline{x})$ is minimized when $\underline{a} = \underline{u}$ and

$$S = \text{Span}\{\underline{u}_1, \dots, \underline{u}_k\}$$

$$d(\underline{a}+S, \underline{x}) = E(\|\underline{x} - \underline{z}\|^2)$$

→ check best approximation affine space notes:-

* Sample principal components:

Case 1 $\underline{\mu}$ and $\underline{\Sigma}$ are known.

$$\underline{\Sigma} = \underline{V} \underline{\Lambda} \underline{V}^T$$

Case 2 $\underline{\mu}$ and $\underline{\Sigma}$ are unknown.

$$\underline{\mu} = \bar{\underline{x}}_n \quad \underline{\Sigma} = \frac{1}{n} \sum_{i=1}^n (\underline{x}_i - \bar{\underline{x}}_n)(\underline{x}_i - \bar{\underline{x}}_n)^T$$

Data matrix $\underline{X}_n$ of size $n \times p$ is

$$\underline{X}_n = \begin{bmatrix} \underline{x}_1^T \\ \underline{x}_2^T \\ \vdots \\ \underline{x}_n^T \end{bmatrix}_{n \times p} \rightarrow 1 \times p$$

$$\underline{S}_n = \underline{G}_n \underline{L}_n \underline{G}_n^T \quad \text{where } \underline{G}_n \underline{G}_n^T = \underline{G}_n^T \underline{G}_n = \underline{I}_p$$

$$\text{and } \underline{L}_n = \text{diag}(\lambda_1, \dots, \lambda_p)$$

with $\lambda_1 > \lambda_2 > \dots > \lambda_p$

The principal component transformation of i th sample.

$$\underline{y}_i^p = \underline{G}_n^T (\underline{x}_i - \bar{\underline{x}}_n)$$

$$Y_n = \begin{bmatrix} \underline{y}_1^T \\ \vdots \\ \underline{y}_n^T \end{bmatrix} = \begin{bmatrix} (\underline{x}_1 - \bar{\underline{x}}_n)' \underline{G}_n \\ \vdots \\ (\underline{x}_n - \bar{\underline{x}}_n)' \underline{G}_n \end{bmatrix} = (\underline{X}_n - \underline{1} \bar{\underline{x}}_n) \underline{G}_n$$

$$\text{i.e. } Y_n = (\underline{X}_n - \underline{1} \bar{\underline{x}}_n) \underline{G}_n$$

$$= \begin{bmatrix} \underline{x}_1' \\ \underline{x}_2' \\ \vdots \\ \underline{x}_n' \end{bmatrix} - \begin{bmatrix} \bar{\underline{x}}_n' \\ \bar{\underline{x}}_n' \\ \vdots \\ \bar{\underline{x}}_n' \end{bmatrix}_{n \times p} \underline{G}_n = \begin{bmatrix} (\underline{x}_1' - \bar{\underline{x}}_n') \\ (\underline{x}_2' - \bar{\underline{x}}_n') \\ \vdots \\ (\underline{x}_n' - \bar{\underline{x}}_n') \end{bmatrix} \underline{G}_n$$

$$= \begin{bmatrix} (\underline{x}_1 - \bar{\underline{x}}_n) \underline{G}_n \\ (\underline{x}_2 - \bar{\underline{x}}_n) \underline{G}_n \\ \vdots \\ (\underline{x}_n - \bar{\underline{x}}_n) \underline{G}_n \end{bmatrix}$$

$$\boxed{\bar{\underline{Y}}_n = 0}$$

$$S_n^Y = \text{diag} (l_1, l_2, \dots, l_p) = \frac{1}{n} \sum_{i=1}^n (\underline{y}_i - \bar{\underline{y}}_n) (\underline{y}_i - \bar{\underline{y}}_n)'$$

→ Among all SLC's of the Random Sample $(\underline{X}_n)_{n \times p}$ $\underline{Y}_1$ has the maximum variance.

$$\text{SLC: Defn. } \omega_i = \underline{a}^T (\underline{x}_i - \bar{\underline{x}}_n) \quad \underline{a} \in \mathbb{R}^p$$

$$\|\underline{a}\| = 1$$

$$\omega_{n \times 1} = [\omega_1, \dots, \omega_n]'_{n \times 1}$$

$$\bar{\omega}_n = \frac{\underline{1}_n' \omega}{n} ; S_{\omega} = \frac{1}{n} \sum_{i=1}^n (\omega_i - \bar{\omega}_n)^2$$

$$= \boxed{\underline{a}^T S_n \underline{a}}$$

intuition:

$$SLC: w_i = \underline{a}^T (\underline{x}_i - \bar{\underline{x}}_n) \quad \begin{array}{l} \nearrow \text{Simple observation} \\ \downarrow \text{Simple mean} \end{array} \quad \begin{array}{l} \underline{a} \in \mathbb{R}^b \\ \|\underline{a}\| = 1 \end{array}$$

① $\rightarrow$ if we vary $\underline{a}$ ^{only}, we will get different SLC for some observation vector.

② $\rightarrow$ if we vary $\underline{x}_i$ ^{only}, we will get different SLC for different observations for a fixed vector $\underline{a}$

for all observation $i=1, \dots, n$

we will get $w_{n \times 1} = [w_1, \dots, w_n]^T_{n \times 1}$ for a fixed $\underline{a} \in \mathbb{R}^b$

$$\begin{aligned} \bar{w}_n &= \frac{\underline{1}^T w}{n} : & S_w &= \frac{1}{n} \sum_{i=1}^n (w_i - \bar{w}_n)^2 \\ & & &= \frac{1}{n} \sum_{i=1}^n \left(\underline{a}^T (\underline{x}_i - \bar{\underline{x}}_n) - \frac{1}{n} \sum_{i=1}^n \underline{a}^T (\underline{x}_i - \bar{\underline{x}}_n) \right)^2 \\ & & &= \underline{a}^T \left(\frac{1}{n} \sum_{i=1}^n (\underline{x}_i - \bar{\underline{x}}_n) (\underline{x}_i - \bar{\underline{x}}_n)^T \right) \underline{a} \\ & & &= \boxed{\underline{a}^T S_n \underline{a}} \end{aligned}$$

Result 1 Among all possible standard linear combination $y_{(1)}$ has the maximum when

$$\underline{y}_{(1)} = \begin{bmatrix} y_{(1)} \\ \vdots \\ y_{(n)} \end{bmatrix} = \begin{bmatrix} \underline{g}_1^T (\underline{x}_1 - \bar{\underline{x}}_n) \\ \vdots \\ \underline{g}_1^T (\underline{x}_n - \bar{\underline{x}}_n) \end{bmatrix}$$

$\underline{g}_1$ is the eigenvector $\rightarrow \lambda_1$ (largest eigenvalue)

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Result-2 Among all possible standardised linear combination which have zero covariance with $y_{(1)}, \dots, y_{(k)}$; $y_{(k+1)}$ has the maximum variance.

Proof:

we have $(X)_{n \times p}$: data matrix.

$$(X)_{n \times p} = \begin{bmatrix} \underline{x}_1^T = (x_{11}, x_{12}, \dots, x_{1p}) \\ \vdots \\ \underline{x}_n^T = (x_{n1}, \dots, x_{np}) \end{bmatrix}_{n \times p}$$

$$(Y)_{n \times p} = \begin{bmatrix} \underline{y}_1^T = (y_{11}, y_{12}, \dots, y_{1p}) \\ \vdots \\ \underline{y}_n^T = (y_{n1}, \dots, y_{np}) \end{bmatrix} = (X_n - \underline{1}_n \bar{x}_n') G_n$$

$G_n = [g_1, \dots, g_p]$ where g_i is the eigenvector corresponding to λ_i (eigenvalue)

$\rightarrow (\lambda_i, g_i)$ is the eigenvalue eigenvector pair of the sample covariance matrix $S_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)(x_i - \bar{x}_n)'$

$S_n \xrightarrow{\text{Spectral decomposition}} G_n L_n G_n^T$; $G_n = [g_1, \dots, g_p]$

$L_n = \text{diag}(\lambda_1, \dots, \lambda_p)$

now, $\text{var}(Y_n) = \text{var}[(X_n - \underline{1}_n \bar{x}_n') G_n]$

$= G_n' \text{var}(X_n) G_n$
 $= G_n' G_n L_n G_n^T G_n = L_n$

[using the estimate]
 $\hat{\text{var}}(Y_n) = S_n$

$$(X_n)_{n \times p} = \begin{bmatrix} X_{(1)} & \dots & X_{(p)} \end{bmatrix} \quad \boxed{X_{(i)}' : n \times 1} \text{ is the } i\text{th principal component}$$

$$\text{var}(X_{(i)}) = \lambda_i^2 \quad (\lambda_i^2 \text{ largest eigen value}).$$

Now:

Result 1 (proof)

$$\text{Since } \lambda_1 > \lambda_2 > \dots > \lambda_n$$

$$\Rightarrow \text{var}(X_{(1)}) \geq \text{var}(X_{(2)}) \geq \dots \geq \text{var}(X_{(n)})$$

Let $X_{(i)}^*$ be any other SLL.

$$X_{(i)}^* = \frac{1}{\sqrt{p-1}} \sum_{j=1}^{p-1} X_{(j)} a_j$$

$$X_{(i)}^* = [X_n - \frac{1}{n} \bar{X}_n \mathbf{1}_n'] \underline{a}_{p \times 1}$$

$$\underline{a}' \underline{a} = 1$$

$$\begin{aligned} \text{var}(X_{(i)}^*) &= \underline{a}' \text{var}(X_n) \underline{a} \\ &= \boxed{\underline{a}' S_n \underline{a}} \quad S_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})(X_i - \bar{X})' \end{aligned}$$

$$\underline{a}' S_n \underline{a} = \frac{1}{n} \sum_{i=1}^n (\underline{a}' (X_i - \bar{X}))^2$$

$$= \underline{a}' \sum_{i=1}^n (\underline{a}' (X_i - \bar{X}))^2 \underline{a}$$

$$= \sum_{i=1}^n \lambda_i (\underline{a}' g_i)^2$$

$$\leq \lambda_1 \sum_{i=1}^n (\underline{a}' g_i)^2 = \lambda_1 \underline{a}' \left(\sum_{i=1}^n g_i g_i' \right) \underline{a}$$

$$= \lambda_1 \sum_{i=1}^n \frac{1}{\lambda_i} (\underline{a}' g_i)^2 \quad \left(\sum_{i=1}^n g_i g_i' = \frac{1}{n} S_n \right)$$

$$= \boxed{\lambda_1} = \text{var}(X_{(1)})$$

$$\Rightarrow \boxed{\text{var}(X_{(i)}^*) \leq \text{var}(X_{(1)})}$$

(14)

Result-2Say $\underline{Z}$ is uncorrelated with $\underline{y}_1, \dots, \underline{y}_k$ 4 $\underline{Z}$ is a SLC i.e. $\underline{Z} = (\underline{X}_n - \underline{1}_n \bar{\underline{x}}_n') \underline{a}_{px}$
n.p.Var($\underline{Z}$) is max. if $\underline{Z} = \underline{y}_{(k+1)}$

$$\text{Cov}(\underline{Z}, \underline{y}_i) \quad \forall i = 1, \dots, k$$

$$= 0$$

$$= \text{Cov}[(\underline{X}_n - \underline{1}_n \bar{\underline{x}}_n') \underline{a}_{px}, (\underline{X}_n - \underline{1}_n \bar{\underline{x}}_n') \underline{g}_i]$$

$$= \underline{a}' \text{Cov}(\underline{X}_n) \underline{g}_i$$

$$= \underline{a}' \underline{S}_n \underline{g}_i \quad \forall i = 1, \dots, k$$

$$= \underline{a}' \underline{G}_n \underline{L}_n \underline{G}_n^T \underline{g}_i = 0$$

$$= \underline{a}' \left(\sum_{j=1}^k \underline{L}_j \underline{g}_j \underline{g}_j' \right) \underline{g}_i = 0$$

$$\left\{ \begin{array}{ll} \text{if } i \neq j & \text{then } \underline{g}_i \underline{g}_j' = 0 \\ \text{if } i = j & \text{or } 1 \end{array} \right\}$$

$$= \underline{a}' \left(\sum_{j=1}^k \underline{L}_j \right) \underline{g}_i = 0$$

$$= \sum_{j=1}^k \underline{L}_j \underline{a}' \underline{g}_j = 0 \Rightarrow \nexists \underline{L}_j / \underline{a}' \underline{g}_j \neq 0$$

— (1)

now

$$\text{var}(\underline{Z}) = \text{var}[(\underline{X}_n - \underline{1}_n \bar{\underline{x}}_n') \underline{a}_{px}]$$

$$= \underline{a}_{px}' \underline{S}_n \underline{a}_{px}$$

$$= \underline{a}' \left(\sum \underline{L}_i \underline{g}_i \underline{g}_i' \right) \underline{a}$$

$$= \sum_{i=1}^p \underline{L}_i \underline{a}' \underline{g}_i \underline{g}_i' \underline{a}$$

$$= \underbrace{\sum_{i=1}^k l_i^0 a' g_i g_i' a}_{0'' \text{ from } \textcircled{1}} + \sum_{j=k+1}^p l_j a' g_j g_j' a$$

$$= \sum_{j=k+1}^p l_j a' g_j g_j' a$$

$$< l_{k+1} \sum_{j=k+1}^p a' g_j g_j' a$$

$$< l_{k+1} \sum_{j=1}^p a' g_j g_j' a$$

$$= l_{k+1} a' \underbrace{\sum_{j=1}^p g_j g_j' a}_{I_p}$$

$$= l_{k+1} a' \underbrace{I_p}_{\rightarrow I_p} a$$

$$= l_{k+1} a' a \xrightarrow{\rightarrow 1}$$

$$= \boxed{l_{k+1}} = \text{var}(X_{(k+1)})$$

$$\boxed{\text{var}(Z) \leq \text{var}(X_{(k+1)})}$$

$$\Rightarrow \text{var}(Z) \text{ is maximum when } \boxed{Z = X_{(k+1)}}$$

Result 3

$$d(a+S, X_n) = \left(\sum_{i=1}^n \|x_i - \hat{x}\|^2 \right)$$

$S \subseteq \mathbb{R}^p$ and if $\dim(S) = k (< p)$ and $a \in \mathbb{R}^p$ then

$d(a+S, X_n)$ is minimized if $a = \bar{x}_n$ and $S = \text{span}\{g_1, \dots, g_k\}$
 $\& G_n = (g_1, \dots, g_k)$

Proof: $\bar{x}_n, x_1, \dots, x_n$ iid $N(\mu, \Sigma)$

$\bar{x}_n$ is the MLE of μ .

$S_n = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x}_n)(x_i - \bar{x}_n)'$ is the MLE of Σ

(16) $\rightarrow (\lambda_1, \dots, \lambda_p)^T$ are the eigen values of Σ .

$\rightarrow (v_1, \dots, v_p)$ are the eigen vectors of Σ corresponds to $(\lambda_1, \dots, \lambda_p)$

Now Assumptions:

① $\rightarrow \Sigma = \sum_{i=1}^p \lambda_i^0 v_i^0 v_i^{0T}$ — All the eigen values are unique,
distinct, real.
(~~not~~ unique)
upto an order

② $\rightarrow$ The first non-zero components v_i are positive.

then under these two assumptions,

$\Rightarrow \Sigma$ has unique spectral decomposition.

$\rightarrow$ Now, $(\lambda_1, \dots, \lambda_p)^T$ is an MLE of $(\lambda_1, \dots, \lambda_p)^T$ when
 $\lambda_1 > \lambda_2 > \dots > \lambda_p$

and g_i is MLE of v_i

when g_i has the first non-zero component positive.

$\forall i = 1, \dots, p$.

$\exists (f_1, \dots, f_p)$ and $(h_1, \dots, h_p)$ [one-one]

$$\text{st } \boxed{f_i(\Sigma) = \lambda_i^0 \text{ \& \& } h_i(\Sigma) = v_i^0}$$